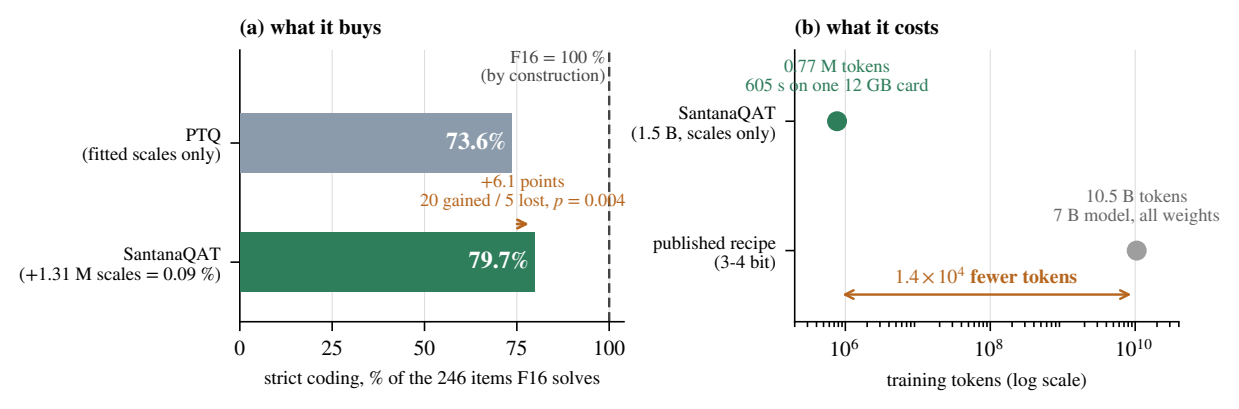
(a) what it buys
F16 = 100 %
(by construction)
PTQ
(fitted scales only)
73.6%
+6.1 points
20 gained / 5 lost, p = 0.004
SantanaQAT
(+1.31 M scales = 0.09 %)
79.7%
0
25
50
75
100
strict coding, % of the 246 items F16 solves
(b) what it costs
0.77 M tokens
605 s on one 12 GB card
SantanaQAT
(1.5 B, scales only)
10.5 B tokens
7 B model, all weights
published recipe
(3-4 bit)
1.4 × 10^4 fewer tokens
10^6
10^8
10^10
training tokens (log scale)